

Younes Mouhib

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French citizen (EU/EFTA, free movement) born 21.08.1998
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ETH Zurich MSc mathematician (August 2026): statistics, stochastic processes, optimisation, constraint solving. Research-first and evidence-driven — record combinatorial bounds, every result independently re-certified; personal momentum research judged only on cost-aware, walk-forward, out-of-sample numbers. Target: graduate quantitative roles — systematic trading • commodities • energy. Available September 2026.

CORE SKILLS

Quantitative: statistics • time-series & momentum signals • probability & stochastic processes • backtesting (walk-forward, out-of-sample, cost-aware) • convex / integer / network optimisation • SAT / constraint solving
Programming: Python (NumPy, pandas, Matplotlib, pytest) • SQL • MATLAB • Excel • Git & CI • $\text{\LaTeX}$
Languages: French (native) • English (fluent) • Arabic (fluent) • Spanish (conversational) • German (learning)

EDUCATION

ETH Zurich — MSc in Mathematics Sept 2023 – Aug 2026 (expected)
Convex Optimisation • Network & Integer Optimisation • Probabilistic Methods • Structural Graph Theory • Algebraic Methods in Combinatorics Zurich

EPFL — BSc in Mathematics Sept 2018 – July 2023
Analysis • probability & measure theory • statistics • numerical mathematics Lausanne

Lycée Masséna — Classes préparatoires MPSI / MP* 2016 – 2018
Intensive mathematics & physics track Nice, France

RESEARCH & PUBLICATIONS

MSc Thesis — record lower bounds for Hales–Jewett numbers 2025 – 2026
ETH Zurich — advisor: Prof. Dr. Lorenz Halbeisen

- Proved an exact symmetry reduction collapsing a $\sim 10^{10}$ -configuration search to a 969-constraint model; SAT search + independent, dependency-free certification set two record bounds ($\text{HJ}(3, 3) \geq 22$, $\text{HJ}(4, 2) \geq 14$) and corrected a published value in the literature.
- Two arXiv preprints — [2607.02226](#) (single author), [2606.22155](#) — all certificates open, CI-verified, reproducible in seconds: [hj-certificates](#).

PROJECTS

Spare-time builds with AI-assisted implementation — listed for the modelling and validation discipline.

Momentum strategy research — cost-aware backtesting (crypto) 2023 – Present
Python • Binance data [repo](#)

- Order-flow-confirmed momentum signals; backtests charge fees & slippage on every fill, next-bar entries, worst-case tie-breaks; walk-forward validation reporting out-of-sample numbers only.

Brigade — multi-agent LLM system, Lean 4 verifier as sole acceptance authority 2024 –
Model-validation architecture: agents propose, an independent formal checker accepts [repo](#)

Hales–Jewett Explorer — interactive thesis companion, exact in-browser counters 2026
[live](#)

TEACHING

Course Instructor (PVK-Leiter), Topology — 4-day intensive, 20+ students June 2026
VSETH, ETH Zurich Zurich

Homework Tutor (grades 9–11) • Student Mentor, EPFL 2020 – 2023
Lausanne

INTERESTS

Chess (SKETHZ, ETH Zurich) • Project Euler & Mathraining • classical & flamenco guitar • martial arts.